

Systems Testing

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Overview

- Introduction to Systems Testing
  - What is system testing?
  - Why do we do system testing?
  - System testing tools
- Types of System Testing
  - Manual systems testing
  - Automatic system testing
  - Continuous integration
- System Testing with F´

Introduction to System Testing

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Systems Testing

- Testing software within the full running system
- Includes flight-like hardware and ground support
- Also called ***integration testing*** and similar to ***regression testing***
- “Test as you fly, fly as you test”

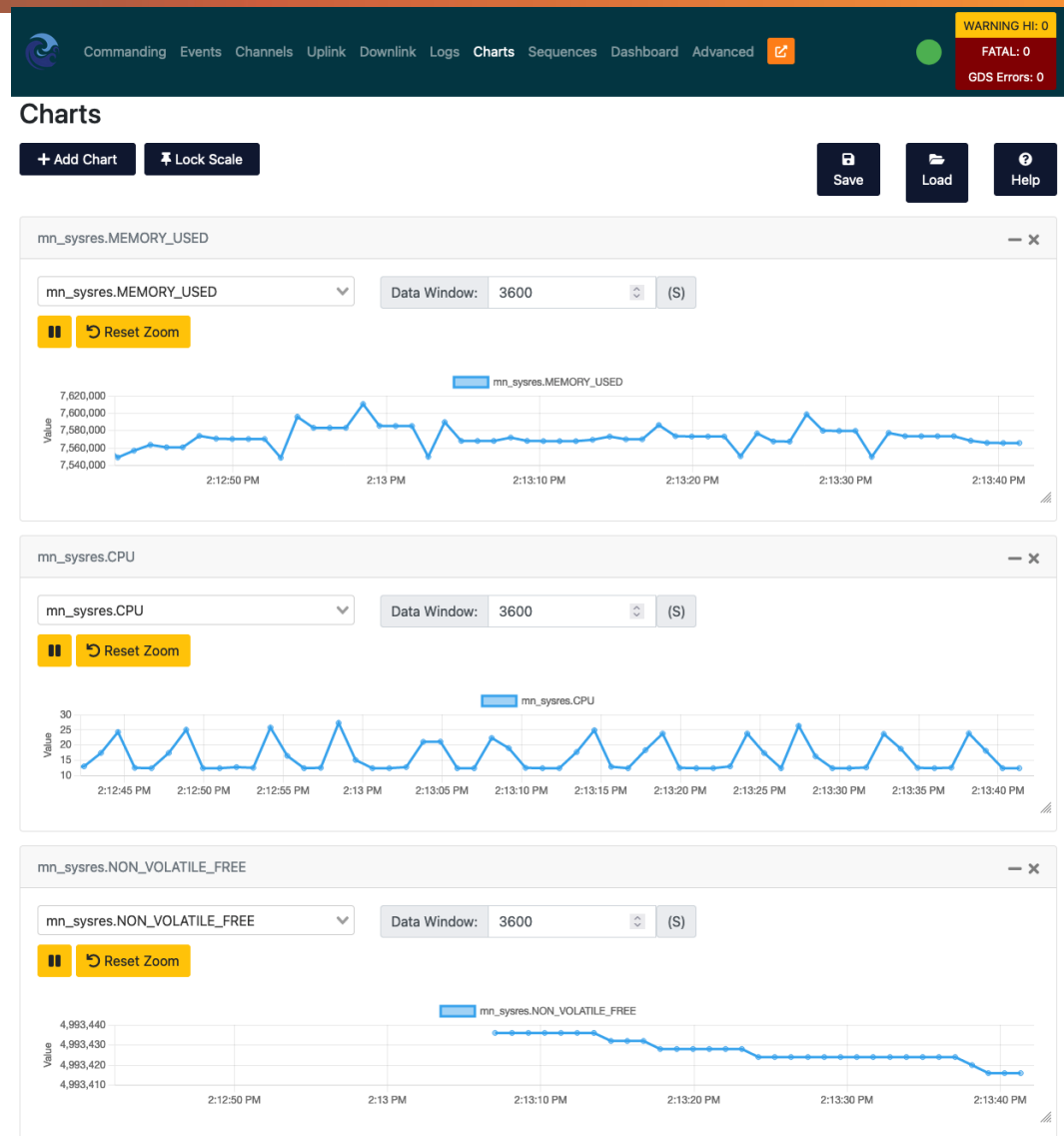


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Why Systems Testing?

- A system is more than the sum of its parts
- Some requirements can only be verified at system level
- Hardware operations need testing on hardware
- “Eat our own dog food”



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Thundering Herd



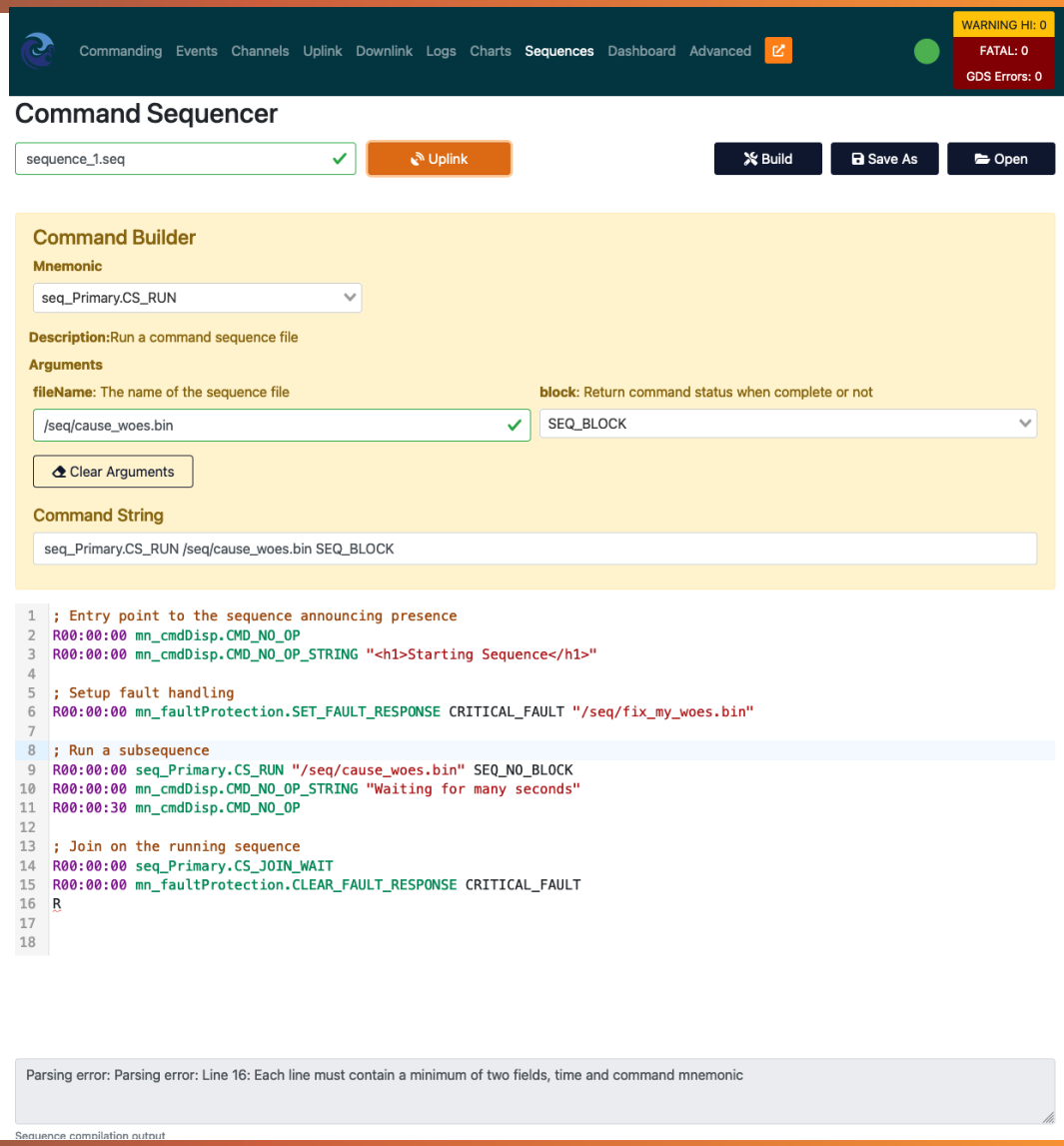
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System Testing Tools

- Ground Data Systems
  - Operate spacecraft from earth
  - May be used to conduct testing
- Testing Frameworks
  - Allow for automated testing
  - Reporting, verification, reproducibility
- Hardware Simulations
  - Allows system testing without physical hardware
- Command Sequences
  - How the flight spacecraft is operated
  - May be used to construct tests



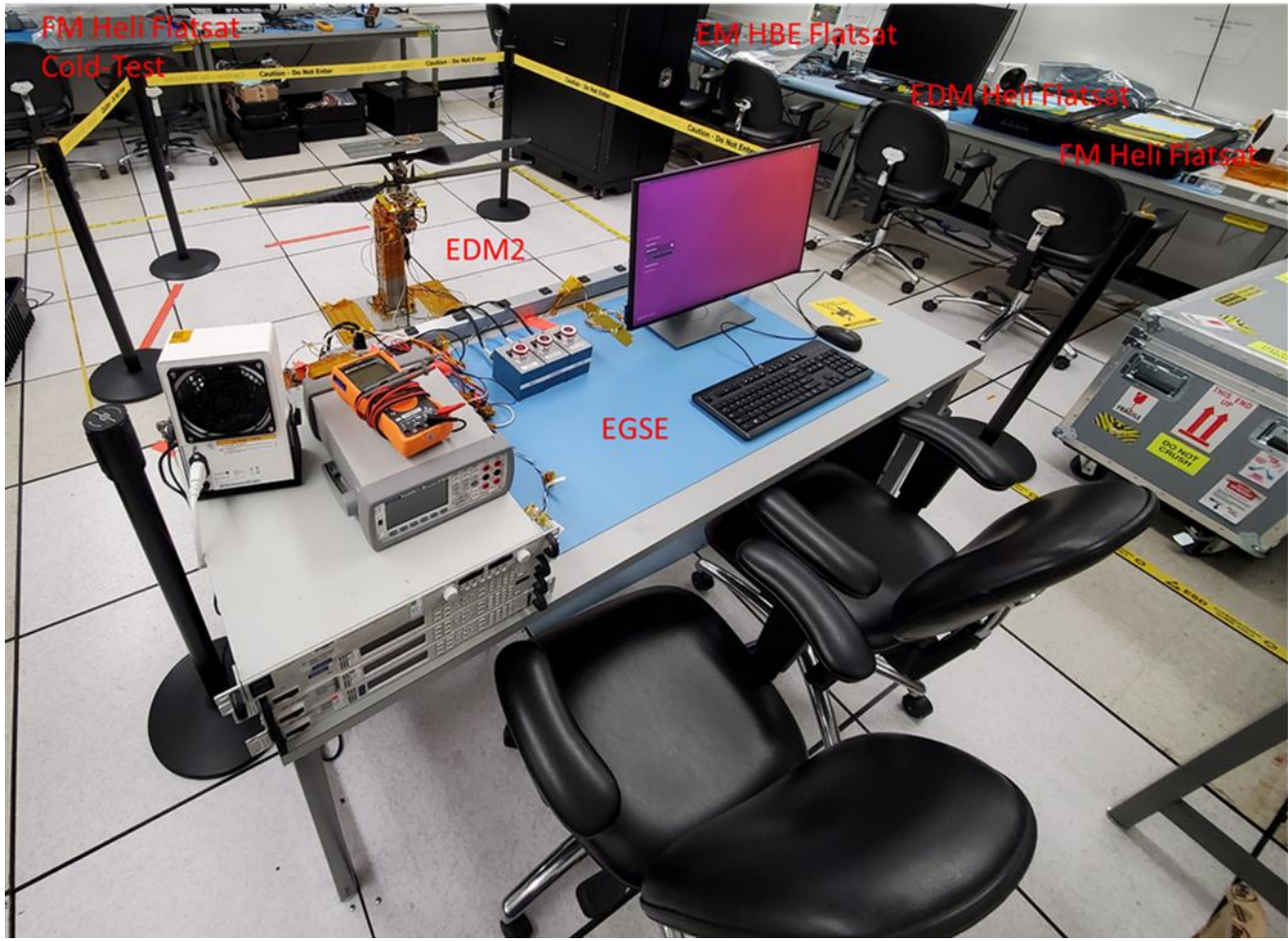
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Types of System Testing

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Manual Systems Testing

- Operating the spacecraft by-hand to verify requirements
- Running sequences and commands and verifying results
- Ground system is used to send commands
- More complete tests often require full test team

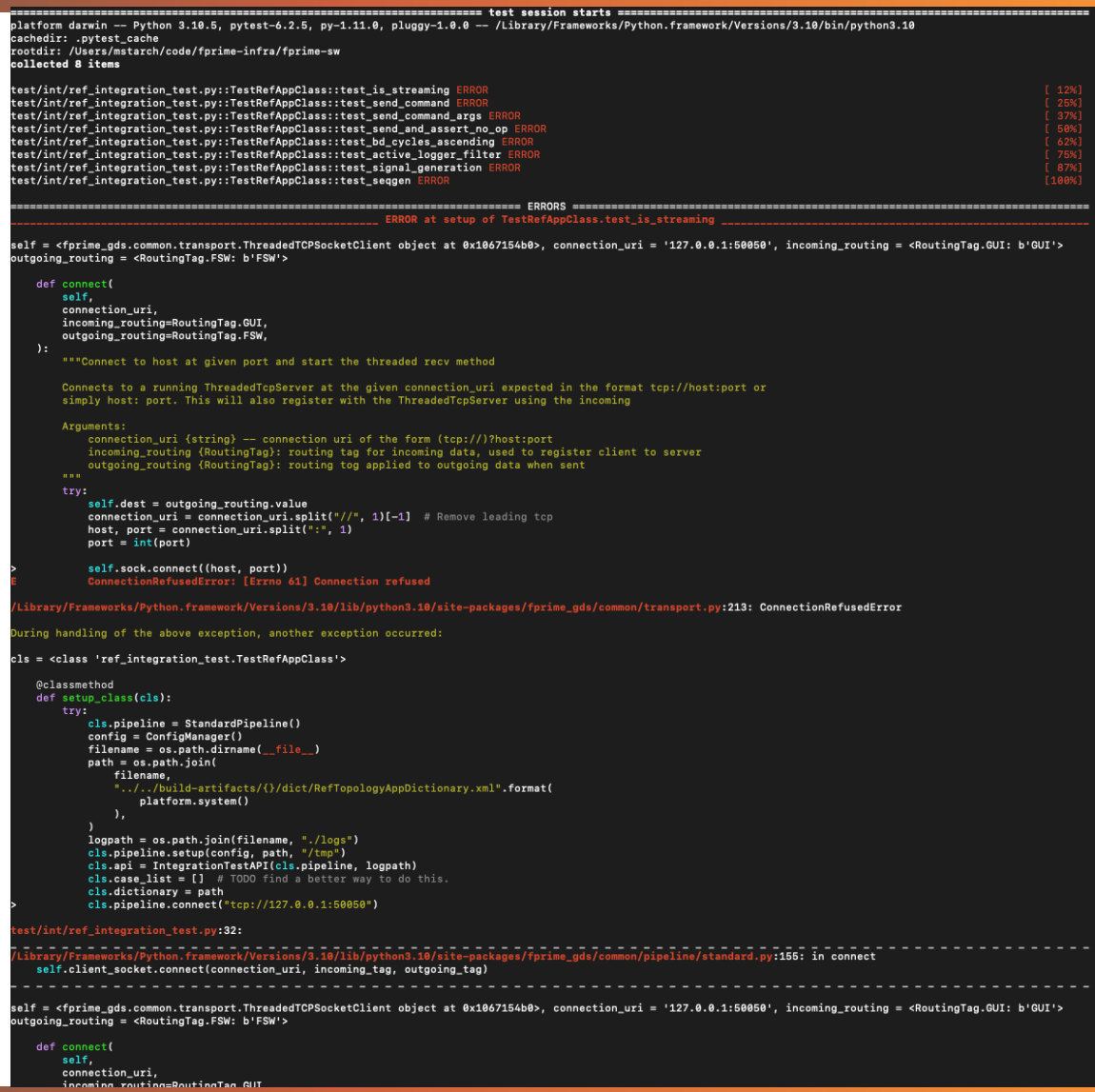


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Automated System Testing

- System testing performed by software
- Uses automated testing support from ground station
- Automatically logs data, results, and verification products
- May not require dedicated test team

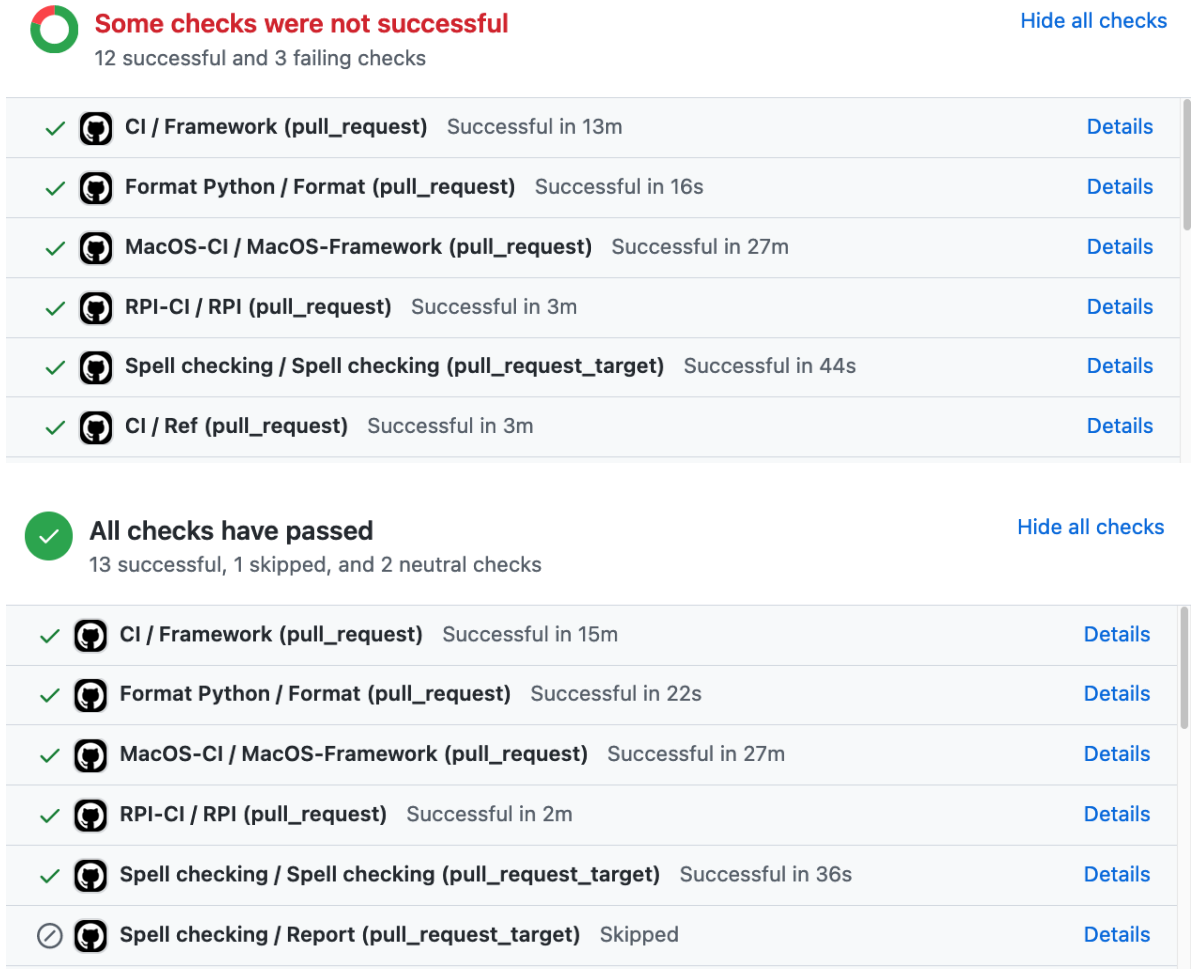


```
============================= test session starts ==============================
platform darwin -- Python 3.10.5, pytest-6.2.5, py-1.11.0, pluggy-1.0.0 -- /Library/Frameworks/Python.framework/Versions/3.10/bin/python3.10
cachedir: .pytest_cache
rootdir: /Users/mstarch/code/fprime-infra/fprime-sw
collected 8 items

test/int/ref_integration_test.py::TestRefAppClass::test_is_streaming ERROR                                                
```

Continuous Integration

- Extension of automatic system testing
- Triggered by software change request or other criteria
- Requires no human interaction
- Implements full regression testing



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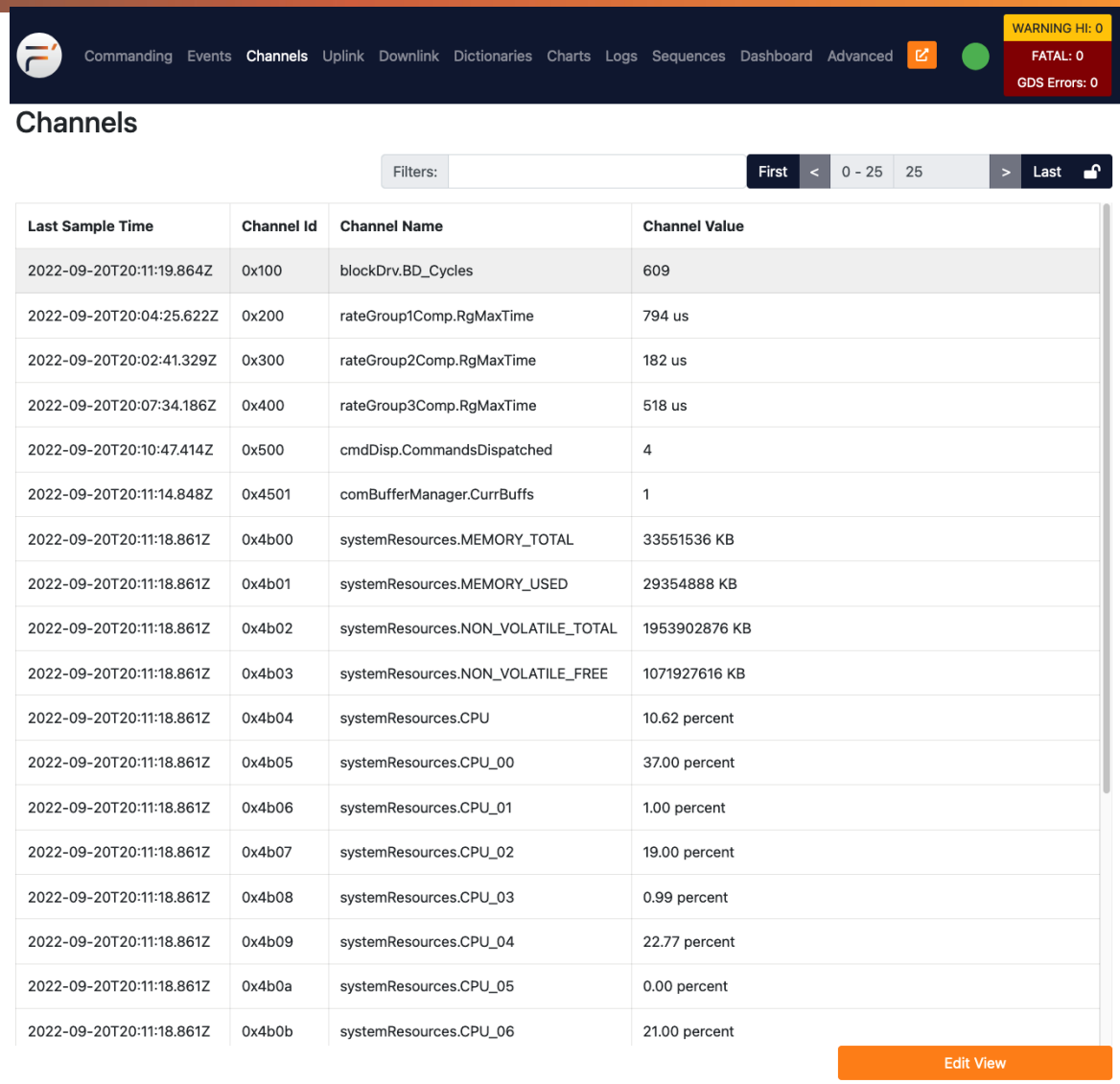


System Testing with F´

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F´ GDS

- Provides GDS capabilities out-of-the-box for F´ projects
- Designed for system testing early in project development
- Enables developers to:
  - Send commands
  - Receive events and telemetry
  - Uplink and downlink files
  - Design and upload sequences
  - Compose custom dashboards



Last Sample Time	Channel Id	Channel Name	Channel Value
2022-09-20T20:11:19.864Z	0x100	blockDrv.BD_Cycles	609
2022-09-20T20:04:25.622Z	0x200	rateGroup1Comp.RgMaxTime	794 us
2022-09-20T20:02:41.329Z	0x300	rateGroup2Comp.RgMaxTime	182 us
2022-09-20T20:07:34.186Z	0x400	rateGroup3Comp.RgMaxTime	518 us
2022-09-20T20:10:47.414Z	0x500	cmdDisp.CommandsDispatched	4
2022-09-20T20:11:14.848Z	0x4501	comBufferManager.CurrBuffs	1
2022-09-20T20:11:18.861Z	0x4b00	systemResources.MEMORY_TOTAL	33551536 KB
2022-09-20T20:11:18.861Z	0x4b01	systemResources.MEMORY_USED	29354888 KB
2022-09-20T20:11:18.861Z	0x4b02	systemResources.NON_VOLATILE_TOTAL	1953902876 KB
2022-09-20T20:11:18.861Z	0x4b03	systemResources.NON_VOLATILE_FREE	1071927616 KB
2022-09-20T20:11:18.861Z	0x4b04	systemResources.CPU	10.62 percent
2022-09-20T20:11:18.861Z	0x4b05	systemResources.CPU_00	37.00 percent
2022-09-20T20:11:18.861Z	0x4b06	systemResources.CPU_01	1.00 percent
2022-09-20T20:11:18.861Z	0x4b07	systemResources.CPU_02	19.00 percent
2022-09-20T20:11:18.861Z	0x4b08	systemResources.CPU_03	0.99 percent
2022-09-20T20:11:18.861Z	0x4b09	systemResources.CPU_04	22.77 percent
2022-09-20T20:11:18.861Z	0x4b0a	systemResources.CPU_05	0.00 percent
2022-09-20T20:11:18.861Z	0x4b0b	systemResources.CPU_06	21.00 percent

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F´ Integration Test Framework

- Framework designed for integration testing with F´ GDS
- Allows users to write automated tests in Python
- Tests dispatch commands and assert on events and telemetry
- Creates excel test logs
- Can be run within continuous integration system

```python
def test_active_logger_filter(self):
    self.set_default_filters()
    try:
        cmd_events = self.api.get_event_pred(severity=EventSeverity.COMMAND)
        actHI_events = self.api.get_event_pred(severity=EventSeverity.ACTIVITY_HI)
        pred = predicates.greater_than(0)
        zero = predicates.equal_to(0)
        # Drain time for dispatch events
        time.sleep(10)

        self.assert_command("cmdDisp.CMD_NO_OP")
        self.assert_command("cmdDisp.CMD_NO_OP")

        time.sleep(0.5)

        self.api.assert_event_count(pred, cmd_events)
        self.api.assert_event_count(pred, actHI_events)

        self.set_event_filter(self.FilterSeverity.COMMAND, False)
        # Drain time for dispatch events
        time.sleep(10)
        self.api.clear_histories()
        self.api.send_command("cmdDisp.CMD_NO_OP")
        self.api.send_command("cmdDisp.CMD_NO_OP")

        time.sleep(0.5)

        self.api.assert_event_count(zero, cmd_events)
        self.api.assert_event_count(pred, actHI_events)
    finally:
        self.set_default_filters()
```


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